

AUTONOMOUS RC CAR WEEK 3 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

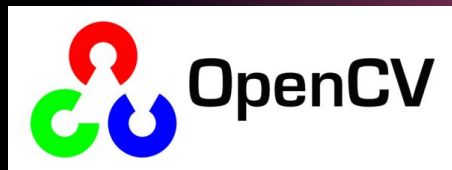
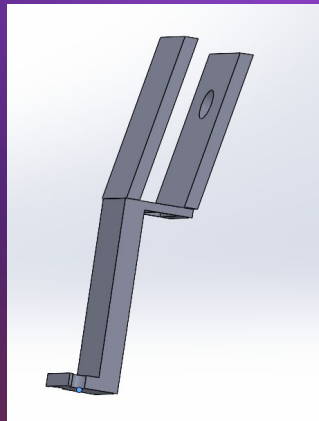
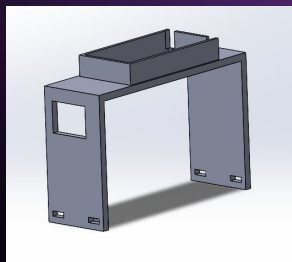
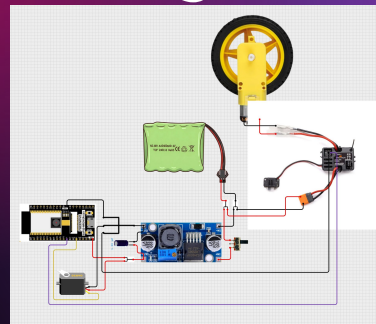
Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

Step 04 (Bonus)

Can it create paths based on given targets?

Progress since our last in person meeting

- Parts arrived, electronics have been wired
- Car can now be controlled from computer via WiFi connection
- Designed and printed 3D prototype mounts
- Tried connecting camera via WiFi but couldn't access Stevens Network
 - Personal hotspot worked but kept crashing due to weak connection



Demo of Car Being Wirelessly Controlled

Next Steps:

- Assemble mounts which are currently being 3D printed
- Get camera operational through the Wifi (need Stevens Wifi Password)
- Use the Camera to detect and avoid obstacles through an AI interface (YOLO) on the laptop
- Start looking into onboard AI processing for the car

THANK YOU

Any questions?